Aravind Ajithkumar

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Profile

Mechanical engineer and M.Sc. Electromobility student at FAU Erlangen-Nürnberg, working where mechanical design meets control. Currently developing a PID-controlled tape-winding tool for a UR10 robot (project thesis at FAPS), building a 6-DOF SO-101 robot arm, and learning Isaac Sim and ROS 2 alongside. Seeking an internship or working-student role on real robotic systems. German A2 and improving. Available by arrangement; open to relocation within Germany.

Education

M.Sc. Electromobility - ACES **10/2022 - present**
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) Erlangen, Germany

- Modules: Machine Learning for Engineers I and II, Programming in C++, Robot Mechanisms and User Interfaces, Integrated Production Systems, Battery Storage Systems

B.Tech. Mechanical Engineering **06/2016 - 09/2020**
Mangalam College of Engineering (APJ Abdul Kalam Technological University) Kerala, India

Experience

Project Thesis - PID-Controlled Robotic Winding Tool (in progress) **04/2026 - present**
Institute FAPS (Factory Automation and Production Systems), FAU Nuremberg, Germany

- Developing a lightweight tape-winding tool for a UR10 robot with ESP32-based PID tension control: spring-loaded dancer roller read by a time-of-flight sensor, stepper motor and driver; PID with speed feedforward and anti-windup, firmware in C++
- Completed the tool design in Fusion 360 and SolidWorks and component selection; printed and checked the FDM prototype parts; hardware delivered
- Simulated the ESP32 firmware and wiring in BenchTwin before the hardware arrived; now integrating the printed tool parts with the electronics. Hardware tests, PID tuning and validation follow

Engineering Intern **02/2020 - 03/2020**
Toyota Industries Engine India Private Limited Bengaluru, India

- Designed two Kaizen countermeasures on the Gemba production floor: an automated clamping lock mechanism (6-8 seconds saved, less worker movement) and an integrated QR-reader/marker-holder tool (2-3 seconds saved per cycle), modelling both mechanisms in SolidWorks. Together with a co-designed head guide, cycle time fell by 17 seconds per engine, which the plant estimated at about 20 additional engines per shift

Projects

SO-101 6-DOF Robot Arm (self-built) **10/2026 - present**
Demo video | GitHub

- Printed the structure (PLA, PETG) on my own Ender 3 S1, integrated six Feetech STS3215 serial bus servos and calibrated all joints with LeRobot
- Programmed pick-and-place of a screwdriver in Python from taught waypoints with interpolated motion; root-caused three faults and documented the build on GitHub. Planned, not yet done: leader-arm teleoperation, ROS 2 integration

KUKA KR210 Palletizing Cell in NVIDIA Isaac Sim (in progress) **2026 - present**
GitHub | Standalone Python, USD, PhysX

- Scripted the cell in Isaac Sim 6.1 (KR210 loaded and verified, cell laid out to scale with pick table and Euro pallet); each stage is a runnable script on GitHub. Motion control is the current stage

Technical Skills

CAD & Design	Fusion 360, SolidWorks, Solid Edge; mechanism and modular assembly design, design for FDM printing
Prototyping	FDM printing (Creality Ender 3 S1, Prusa XL; PLA, PETG), PrusaSlicer, Bambu Studio; workshop basics: drilling, sawing, grinding, soldering, crimping
Control & Embedded	PID design with feedforward and anti-windup (concept stage, hardware tests pending), ESP32, Arduino, stepper drives, time-of-flight sensing
Robotics & Simulation	NVIDIA Isaac Sim 6.1, LeRobot, 2-link inverse kinematics; ROS 2 Humble and Gazebo (learning)
Programming & Tools	Python (beginner to intermediate), C++, MATLAB, Git/GitHub, Linux (Ubuntu, WSL2), LaTeX
Lean Manufacturing	Standardized work, muda analysis, kaizen, Toyota Production System
Languages	English (C1), German (A2, course in progress), Malayalam (native)

Achievements

- **Indian patent application No. 201941015063 (2019)** - aquarium filtration system, one of six co-inventors; proposed the drizzle-type aeration design and built the working prototype
- **SPS Makeathon 2024, Nuremberg** - built a motion bot in a team with Baumüller engineers under time pressure
- **SIEMENS x BEST Engineering Competition 2025** - team-based engineering challenge